# DAVID LYDSTON

**Principal Product Manager | Customer Data Platforms, Machine Learning & Personalization, Experimentation & AI**

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## SUMMARY

Principal Product Manager who turns customer data into products. I am the product manager for **analysis of experiments in Microsoft 365 and Copilot** (Microsoft's generative artificial intelligence (AI) assistant; about **$95 billion in annual revenue**, fiscal year 2025), owning the data platform that turns raw usage events into governed user-, organization-, and device-level data for **3,000+ A/B experiments a month across 30+ products** (**75%+ compute reduction**). Before product management, I built **machine-learning (ML) datasets and models from granular customer events for loyalty personalization, recommendations, and fraud detection** as a data scientist for Xbox Live Rewards, Microsoft's loyalty program serving 3 million gamers (which I helped launch). I also worked as a SQL Server data engineer and a **biostatistician** on clinical trials for U.S. Food and Drug Administration (FDA) submissions. I know what ML scientists need from a data platform because I used to be one of its customers.

## CORE SKILLS

- **Data products & modeling:** strategy and roadmaps for shared data platforms; event- and entity-level datasets (user, organization, device); governed metric catalogs; identity
- **Machine learning & personalization:** predictive analytics, customer segmentation (expectation-maximization (EM) clustering), Bayesian forecasting, time-series regression, product recommendation engines, value-based loyalty targeting, campaign return-on-investment (ROI) analysis
- **Experimentation & statistics:** A/B testing at scale, variance reduction (CUPED: Controlled-experiment Using Pre-Experiment Data), sequential testing, survival analysis, mixed-effects models; SAS
- **Data engineering & SQL:** Structured Query Language (SQL) and SQL Server (former senior database administrator; Microsoft-certified), Apache Spark, Microsoft Cosmos (internal exabyte-scale data lake), Kusto / Azure Data Explorer, data warehousing, extract-transform-load (ETL), Online Analytical Processing (OLAP)
- **AI & large language model (LLM) products:** AI agents, retrieval over historical data, LLM-generated metadata, AI usage and cost metrics

## PROFESSIONAL EXPERIENCE

### Principal Product Manager | Microsoft IDEAS

Aug 2024 – Present

*IDEAS is the central data, analytics, and experimentation organization for Microsoft's Experiences + Devices division (Microsoft 365, Copilot, Teams, Windows). Expedite is IDEAS's experiment decision portal, where product teams and leaders review A/B test results and make ship / no-ship calls.*

- Own vision, strategy, and roadmap for Microsoft's largest cross-organization **experiment scorecard platform** (statistical results for each A/B test): **30+ products, 3,000+ experiments a month**, driving Microsoft 365 and Copilot launch decisions.
- Required every metric to come from **UDM (Unified Data Model)**, Microsoft 365's governed shared data model, and added **experiment-assignment data for 30+ products** to it so any experiment joins to any governed user, organization, or device metric.
- Grew coverage **~50x** to **85%+ of all Experiences + Devices experiments**, cut results delivery from **3 days to 36 hours**, and reduced compute **75%+** with fast metrics and lean scorecards.
- Relaunched Expedite as Copilot's **ship / no-ship gate** (7,838 active Copilot experiments). In the past year our scorecards flagged significant changes in **346 Copilot features** tested on up to **197 million users**: **132 that hurt Copilot chat weekly active users or actions per daily active user were held back or stopped before full release**; **58 regression-free wins shipped**.
- Serve **data scientists as core customers**, translating their needs into product specifications for **anytime-valid experiment monitoring** (reliable repeated checks) and **Controlled-experiment Using Pre-Experiment Data (CUPED) variance reduction**, including historical-data requirements, validation workflows, and false-positive controls.
- Partner with data science on **experiment-analyzer**, an **AI assistant for interpreting experiments**; own product requirements for user inputs, metric coverage, acceptance testing, and auditable decision reports.
- Aligned six partner teams on a shared **customer lifetime value (LTV) scorecard**, standardizing measurement across **hundreds of features**.
- Product leader for a **24-person team** (15 engineers, 5 data scientists, 4 product managers) through reorganizations; kept sensitive customer data off unsecured scorecards to protect privacy.

### Principal Product Manager | Microsoft Office Growth

Nov 2022 – Aug 2024

*Growth and experimentation for Office apps (Word, Excel, PowerPoint, OneNote) on web, desktop, and mobile.*

- Prioritized an **AI usage (Copilot token-consumption) scorecard** across all Office experiments and rollouts; it caught a **7% drop in Copilot top-of-funnel engagement** in OneNote ($p < 0.001$) that would otherwise have shipped unnoticed.
- Co-authored the fiscal year 2025 experimentation strategy, adopted by senior leadership as the organization's vision.

- Reached **Microsoft's lowest sample ratio mismatch rate** (a data-quality check for broken traffic splits) via an alert-to-bug pathway.

**Principal Product Manager | Microsoft Analysis & Experimentation** Sep 2020 – Nov 2022

*Analysis & Experimentation runs ExP, Microsoft's company-wide A/B testing platform used by Bing, Microsoft 365, Windows, Azure, and Dynamics 365. Bing is part of Microsoft's search and news advertising business (about $14B annual revenue).*

- Co-led a **100,000-token daily compute reduction** (Cosmos big data processing) with Bing, cutting Bing's **ship-decision time from six days to two**.
- Delivered **$7M in scorecard compute-cost savings** with Bing through three platform changes and a data science-led optimization.
- Negotiated cost sharing with Bing to fund the **migration from Cosmos to Apache Spark**.

**Senior Program Manager | Microsoft Analysis & Experimentation** May 2018 – Sep 2020

- Cut the Bing ship-decision scorecard SLA from ~13 hours to **under 10 hours** by shipping a caching layer for Bing web search jobs (**70% fewer processing-node hours**).
- Authored specs for **audience targeting filters** that let teams target experiments to customer segments, and migrated the real-time experiment assignment service under a fixed deadline.

**Senior Program Manager | Microsoft** Feb 2016 – May 2018

*Digital analytics (Adobe Analytics) for Microsoft Store, Windows, and MSN.*

- Built a **cohort analysis proof of concept** for senior leadership that quantified the revenue impact of e-commerce entitlement failures (customers unable to access purchases), driving investment in Microsoft Store storefront optimization.
- Cut Adobe Analytics sampling volume 50% (**$60K+/month saved**) and negotiated terms positioning Microsoft to **save $140K/month**.

**Data Scientist, Client Services & Strategy | HelloWorld (formerly ePrize)** May 2012 – Jan 2016

*HelloWorld (now part of Merkle) runs digital promotions and loyalty programs for global brands. My main client was Microsoft's Xbox Live Rewards (rewards.xbox.com), which rewarded Xbox gamers for purchases and engagement. Joined when ePrize acquired Apollo.*

- Analyzed and cleaned granular transaction and behavioral events from Xbox Live Rewards, preparing machine-learning datasets and building models for **classification, personalized recommendations, and fraud detection** for a loyalty program serving **3 million members**.
- Used ML to design the levels and qualifying thresholds of **MyVIP**, Xbox Live Rewards' membership tier program, and measured its impact versus a stratified control group in fiscal year 2015: **300% more points earned** and **$136,840 in incremental revenue**. ML-set thresholds and earning amounts across all Xbox Live Rewards campaigns added **$284,229**.
- Built a **product recommendation engine** for cross-sell and cross-engagement offers.

**Database & Systems Engineer (contract to Microsoft) | Apollo Data Technologies** Jul 2008 – May 2012

- Member of the team that **launched Xbox Live Rewards in December 2010**. Designed the data platform behind it (processing, reporting, and analysis layers), scaled to **250,000 new members a day**, a 5-terabyte processing platform, and 900 gigabytes of OLAP cubes; cut processing times **30%** and data-layer size by **2.5 terabytes**.

**Senior Database Administrator & Statistics Analyst | Bio-Rad Laboratories** Aug 2005 – Jul 2008

*Medical diagnostics manufacturer; role combined clinical data management with biostatistics.*

- Performed biostatistics in SAS for clinical studies supporting **FDA submissions**: instrument precision (mixed-effects models), product survival (Kaplan-Meier), and sensitivity and specificity with 95% confidence intervals.
- Built a **clinical trial data management system** with automated validation, replacing per-trial spreadsheets; recalled in 2013 to extend it.

**BioAnalyst | Poniard Pharmaceuticals** Nov 2004 – Aug 2005

- Designed the data capture system for a **pivotal Phase III oncology trial** and analyzed it in SAS (survival analysis; hundreds of result tables for **FDA submission**). Earlier, as Systems Analyst III at the **University of Miami** (1999–2004), led SQL Server clinical-research data systems, including handheld data capture for multinational AIDS research.

## EDUCATION & PUBLICATIONS

**Master of Science, Information Systems Management**, Seattle Pacific University, 2007 | **Master of Arts, Behavioral Psychology**, San Diego State University, 1999 | **Bachelor of Arts, Psychology**, Western Washington University, 1997

- Co-author of **8 peer-reviewed publications** (2001–2010; 200+ citations) in journals including AIDS Care and Journal of Psychosomatic Research; applied data science article (CMO.com, 2015) and talk (SQL Saturday, 2014).